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(1)

Module 3

Ordinary Differential Equations (ODE)

Homogeneous linear ODE with constant coefficients

The general form of a second order homogeneous linear ODE with constant coefficients

is $\frac{d^2 y}{dx^2} + a \frac{dy}{dx} + by = 0$ or $y'' + ay' + by = 0$

eg: $y'' + 3y' + 2y = 0$

Superposition principle of homogeneous linear ODE

If y_1 and y_2 are two solutions of a second order homogeneous linear ODE, then their linear combination, $y = c_1 y_1 + c_2 y_2$ will also be a solution.

eg: Consider the equation $y'' + y = 0$

Here $y_1 = \cos x$ is a solution [Since $y_1' = -\sin x$
 $y_1'' = -\cos x$
 $y_1'' + y_1 = -\cos x + \cos x = 0$]

Similarly $y_2 = \sin x$ is a solution [Since $y_2' = \cos x$
 $y_2'' = -\sin x$
 $y_2'' + y_2 = -\sin x + \sin x = 0$]

Then $y = 2\cos x + 3\sin x$ is also a solution.

[Since $y' = -2\sin x + 3\cos x$; $y'' = -2\cos x - 3\sin x$

$$y'' + y = -2\cos x - 3\sin x + 2\cos x + 3\sin x = 0$$

Note: This principle does not hold for a non homogeneous linear or non linear ODE.

(2)

Initial value problem (i.v.p)

For a second order homogeneous linear ODE, i.v.p consists of two initial conditions, $y(x_0) = k_0$ and $y'(x_0) = k_1$.

These conditions are used to determine the arbitrary constants c_1 and c_2 in the general solution $y = c_1 y_1 + c_2 y_2$. The solution thus obtained is called particular solution.

eg: Consider $y'' + y = 0$, $y(0) = 3$, $y'(0) = -0.5$
Solve the initial value problem.

Soln:- General solution is $y = c_1 \cos x + c_2 \sin x$

$$y(0) = 3 \Rightarrow c_1 \cos 0 + c_2 \sin 0 = 3 \Rightarrow c_1 = 3$$

$$y'(0) = -0.5 \Rightarrow \left[-c_1 \sin x + c_2 \cos x \right]_{x=0} = -0.5$$

$$\Rightarrow c_2 = -0.5$$

$\therefore y = 3 \cos x - 0.5 \sin x$ is the particular solution

General Solution and Basis

The general solution of a second order homogeneous linear ODE will be in the form

$y = c_1 y_1 + c_2 y_2$ where y_1 and y_2 are linearly independent solutions of the equation.

The set $\{y_1, y_2\}$ of linearly independent solutions is called a basis.

(3)

To check linear dependence and independence

Two solutions y_1 and y_2 are said to be linearly dependent if $y_1 = ky_2$ or

$\frac{y_1}{y_2} = k$, constant. Otherwise, linearly independent.

eg: Consider $y'' + y = 0$

Here $y_1 = \cos x$ and $y_2 = \sin x$ are two solutions and they are linearly independent because

$$\frac{y_1}{y_2} = \frac{\cos x}{\sin x} = \cot x \neq \text{constant}.$$

Hence $y = c_1 \cos x + c_2 \sin x$ is the general solution and $\{\cos x, \sin x\}$ forms a basis of the equation $y'' + y = 0$

1. Verify by substitution that $y_1 = e^x$ and $y_2 = e^{-x}$ are solutions of the equation $y'' - y = 0$ and are linearly independent. Then solve the i.v.p, $y'' - y = 0$, $y(0) = 6$, $y'(0) = -2$.

Soln : $y_1 = e^x$; $y_1' = e^x$; $y_1'' = e^x$

$$y_1'' - y_1 = e^x - e^x = \underline{0}$$

$$y_2 = e^{-x}, \quad y_2' = -e^{-x}, \quad y_2'' = e^{-x}$$

$$y_2'' - y_2 = e^{-x} - e^{-x} = 0$$

Therefore $y_1 = e^x$ and $y_2 = e^{-x}$ are solutions.

(4)

Also they are linearly independent because $\frac{y_1}{y_2} = \frac{e^x}{e^{-x}} = e^{2x} \neq \text{constant}$.

Therefore general solution is $y = c_1 e^x + c_2 e^{-x}$

$$\text{Given } y(0) = 6 \Rightarrow c_1 e^0 + c_2 e^0 = 6 \Rightarrow c_1 + c_2 = 6 \quad \text{--- (1)}$$

$$y'(0) = -2 \Rightarrow [c_1 e^x - c_2 e^{-x}]_{x=0} = -2 \Rightarrow c_1 - c_2 = -2 \quad \text{--- (2)}$$

Solving (1) and (2) $c_1 = 2$; $c_2 = 4$

$\therefore y = 2e^x + 4e^{-x}$ is the particular solution.

To check linear dependence and independence using Wronskian

Two solutions y_1 and y_2 are linearly dependent if and only if their Wronskian

$$W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = 0$$

1. Show that $y_1 = e^{-0.5x}$ and $y_2 = e^{-2.5x}$ are linearly independent using Wronskian.

$$\begin{aligned} \text{Soln : } W &= \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^{-0.5x} & e^{-2.5x} \\ -0.5e^{-0.5x} & -2.5e^{-2.5x} \end{vmatrix} \\ &= -2.5e^{-3x} + 0.5e^{-3x} = -2e^{-3x} \neq 0 \end{aligned}$$

Hence y_1 and y_2 are linearly independent.

(5)

2. Show that $y_1 = x^2$, $y_2 = 5x$, $y_3 = 2x$ are linearly dependent.

Soln : $W = \begin{vmatrix} x^2 & 5x & 2x \\ 2x & 5 & 2 \\ 2 & 0 & 0 \end{vmatrix} = x^2(0) - 5x(-4) + 2x(-10)$
 $= 20x - 20x = 0$

Hence y_1, y_2, y_3 are linearly dependent.

Problems

I) Verify that given functions are linearly independent and form a basis of solution of the given ODE. Solve the IVP.

1) $y'' + 9y = 0$; $y(0) = 2$; $y'(0) = -1$
 $\cos 3x, \sin 3x$

2) $y'' + 2y' + y = 0$; $y(0) = 2$; $y'(0) = -1$
 e^{-x}, xe^{-x}

II) Check whether the given solutions are linearly dependent or not.

1) $y_1 = x$, $y_2 = x^2$, $y_3 = x^3$

2) $1, x, x^2, x^3$

(6)

To solve a homogeneous linear ODE with constant coefficients

An n^{th} order homogeneous linear ODE with constant coefficients is of the form

$$\frac{d^n y}{dx^n} + a_1 \frac{d^{n-1} y}{dx^{n-1}} + a_2 \frac{d^{n-2} y}{dx^{n-2}} + \dots + a_{n-1} \frac{dy}{dx} + a_n y = 0 \quad \text{--- (1)}$$

Usually we write $\frac{d}{dx} = D$, $\frac{d^2}{dx^2} = D^2$,

Therefore eq(1) can be written as

$$(D^n + a_1 D^{n-1} + a_2 D^{n-2} + \dots + a_{n-1} D + a_n) y = 0 \\ \Rightarrow f(D) y = 0$$

Auxiliary equation or characteristic equation

The equation $f(D) = 0$ is called auxiliary equation or characteristic equation.

Let $D = m_1, m_2, \dots, m_n$ be the roots of A.E

Case I : If all the roots are real and distinct then the general soln, $y = c_1 e^{m_1 x} + c_2 e^{m_2 x} + \dots + c_n e^{m_n x}$

Case II : If two roots are equal, $m_1 = m_2$, then $y = (c_1 x + c_2) e^{m_1 x} + c_3 e^{m_3 x} + \dots + c_n e^{m_n x}$.

If three roots are equal, $m_1 = m_2 = m_3$, then $y = (c_1 x^2 + c_2 x + c_3) e^{m_1 x} + c_4 e^{m_4 x} + \dots + c_n e^{m_n x}$

Case III : If the roots are imaginary, $m_1, m_2 = \alpha \pm i\beta$ then $y = e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x) + c_3 e^{m_3 x} + \dots + c_n e^{m_n x}$

(7)

Case IV : If two pairs of imaginary roots are equal, $m_1, m_2 = \alpha \pm i\beta$ then m_3, m_4

$$y = e^{\alpha x} \left[(c_1 x + c_2) \cos \beta x + (c_3 x + c_4) \sin \beta x \right] + c_5 e^{m_3 x} + \dots + c_n e^{m_n x}$$

1. Solve $y'' + 5y' + 6y = 0$

Soln : $\frac{d^2 y}{dx^2} + 5 \frac{dy}{dx} + 6y = 0 \Rightarrow (D^2 + 5D + 6)y = 0$

Auxiliary eqn (A.E) is $D^2 + 5D + 6 = 0$

$$\Rightarrow (D+3)(D+2) = 0$$

$$\Rightarrow D = -3, -2 \quad (\text{roots are real \& distinct})$$

$\therefore$ General solution is $y = \underline{\underline{c_1 e^{-3x} + c_2 e^{-2x}}}$

2. Find the general solution of

$$\frac{d^3 y}{dx^3} - 7 \frac{dy}{dx} - 6y = 0$$

Soln : The given equation can be written as

$$(D^3 - 7D - 6)y = 0$$

A.E is $D^3 - 7D - 6 = 0$

$$\Rightarrow D = -1, -2, 3$$

Roots are real and distinct

$\therefore y = \underline{\underline{c_1 e^{-x} + c_2 e^{-2x} + c_3 e^{3x}}}$

$D = -1$ is one root

$$\begin{array}{r|rrrr} -1 & 1 & 0 & -7 & -6 \\ & & -1 & 1 & 6 \\ \hline & 1 & -1 & -6 & 0 \end{array}$$

$$\Rightarrow D^2 - D - 6 = 0$$

$$\Rightarrow (D-3)(D+2) = 0$$

$$\Rightarrow D = 3, -2$$

(8)

3. Solve $y''' - 4y'' + 4y' = 0$

Soln : $(D^3 - 4D^2 + 4D)y = 0$

A.E is $D^3 - 4D^2 + 4D = 0 \Rightarrow D(D^2 - 4D + 4) = 0$

$$\Rightarrow D(D-2)^2 = 0$$

$$\Rightarrow D = 0, 2, 2$$

One root is real & distinct, two roots are real & equal.

$$\therefore y = c_1 e^{0x} + (c_2 x + c_3) e^{2x} = c_1 + \underline{(c_2 x + c_3) e^{2x}}$$

4. Solve $\frac{d^4 y}{dx^4} + 13 \frac{d^2 y}{dx^2} + 36y = 0$

Soln : The given equation is $(D^4 + 13D^2 + 36)y = 0$

A.E is $D^4 + 13D^2 + 36 = 0$

$$\Rightarrow D = \pm 3i, \pm 2i$$

Roots are imaginary

$$\therefore y = e^{0x} (c_1 \cos 3x + c_2 \sin 3x)$$

$$+ e^{0x} (c_3 \cos 2x + c_4 \sin 2x)$$

Put $D^2 = u$
 $\therefore u^2 + 13u + 36 = 0$
 $(u+9)(u+4) = 0$
 $u = -9 \quad u = -4$
 $\Rightarrow D^2 = -9 \quad D^2 = -4$
 $\Rightarrow D = \pm 3i \quad D = \pm 2i$

5. Solve $\frac{d^2 y}{dx^2} - 4 \frac{dy}{dx} + y = 0$

Soln : A.E is $D^2 - 4D + 1 = 0$

$$\Rightarrow D = \frac{4 \pm \sqrt{16-4}}{2} = \frac{4 \pm 2\sqrt{3}}{2} = 2 \pm \sqrt{3}$$

Roots are real & distinct

$$\therefore y = c_1 e^{(2+\sqrt{3})x} + c_2 e^{(2-\sqrt{3})x}$$

(9)

6. Find the general solution of
$$\frac{d^2x}{dt^2} + 6\frac{dx}{dt} + 9x = 0$$

Soln : The given equation is

$$(D^2 + 6D + 9)x = 0 \quad \text{where } D = \frac{d}{dt}$$

$$\text{A.E. is } D^2 + 6D + 9 = 0 \Rightarrow (D+3)(D+3) = 0 \\ \Rightarrow D = -3, -3$$

Roots are real & equal.

$\therefore$ General solution is $x = \underline{(c_1 t + c_2) e^{-3t}}$

7. Solve $y''' - 3y'' + 3y' - y = 0$

Soln : $(D^3 - 3D^2 + 3D - 1)y = 0$

A.E. is $D^3 - 3D^2 + 3D - 1 = 0$

$$\Rightarrow (D-1)^3 = 0 \Rightarrow D = 1, 1, 1 \text{ (equal roots)}$$

$$\therefore y = \underline{(c_1 x^2 + c_2 x + c_3) e^x}$$

8. Solve the I.V.P $y'' + 4y' + 29y = 0$;

$$y(0) = 0 ; \quad y'(0) = 15$$

Soln : A.E. is $D^2 + 4D + 29 = 0$

$$D = \frac{-4 \pm \sqrt{16 - 116}}{2} = -2 \pm 5i$$

$$\therefore y = e^{-2x} (c_1 \cos 5x + c_2 \sin 5x)$$

$$y(0) = 0 \Rightarrow e^0 (c_1 + 0) = 0 \Rightarrow \underline{c_1 = 0}$$

$$y'(0) = 15 \Rightarrow \left[e^{-2x} (-5c_1 \sin 5x + 5c_2 \cos 5x) - 2e^{-2x} (c_1 \cos 5x + c_2 \sin 5x) \right]_{x=0} = 15$$

$$\Rightarrow 5c_2 - 2c_1 = 15 \Rightarrow 5c_2 = 15 \Rightarrow \underline{c_2 = 3}$$

(10)

$$\therefore y = \underline{\underline{3e^{-2x} \sin 5x}}$$

9. Solve $\frac{d^4 x}{dt^4} + 4x = 0$

Soln : The given equation is $(D^4 + 4)x = 0$ when $D = \frac{d}{dt}$

A.E is $D^4 + 4 = 0$

$$(D^2 + 2)^2 - (2D)^2 = 0$$

$$(D^2 + 2D + 2)(D^2 - 2D + 2) = 0$$

$$\because a^2 - b^2 = (a+b)(a-b)$$

$$\Rightarrow D = \frac{-2 \pm \sqrt{-4}}{2} \quad D = \frac{2 \pm \sqrt{-4}}{2}$$

$$\Rightarrow D = -1 \pm i \quad D = 1 \pm i$$

$$\therefore x = \underline{\underline{e^{-t} (c_1 \cos t + c_2 \sin t) + e^t (c_3 \cos t + c_4 \sin t)}}$$

10. Solve $\left\{ (D^2 + 1)^3 (D^2 + D + 1)^2 \right\} y = 0$

Soln : A.E is $(D^2 + 1)^3 (D^2 + D + 1)^2 = 0$

$$\Rightarrow (D^2 + 1)^3 = 0 \quad \text{and} \quad (D^2 + D + 1)^2 = 0$$

$$\Rightarrow (D^2 + 1)(D^2 + 1)(D^2 + 1) = 0 \quad (D^2 + D + 1)(D^2 + D + 1) = 0$$

$$\Rightarrow D = \pm i, \pm i, \pm i \quad D = \frac{-1 \pm \sqrt{-3}}{2}, \frac{-1 \pm \sqrt{-3}}{2}$$
$$= \frac{-1 \pm i\sqrt{3}}{2}, \frac{-1 \pm i\sqrt{3}}{2}$$

$$\therefore y = (c_1 x^2 + c_2 x + c_3) \cos x + (c_4 x^2 + c_5 x + c_6) \sin x$$
$$+ \underline{\underline{e^{-\frac{1}{2}x} \left[(c_7 x + c_8) \cos\left(\frac{\sqrt{3}}{2}x\right) + (c_9 x + c_{10}) \sin\left(\frac{\sqrt{3}}{2}x\right) \right]}}$$

(11)

To find the homogeneous linear ODE if the solutions are given

I) Find a second order homogeneous linear ODE for which the given functions are the solutions.

i) e^{-x} , e^{-2x}

Soln : General solution is $y = c_1 e^{-x} + c_2 e^{-2x}$

$\therefore$ Roots of the A.E are -1 and -2 .

$\therefore$ A.E is $(D+1)(D+2)=0 \Rightarrow D^2+3D+2=0$

$\therefore$ The equation is $(D^2+3D+2)y=0$ or

$$\frac{d^2 y}{dx^2} + 3 \frac{dy}{dx} + 2y = 0$$

ii) $\cos 5x$, $\sin 5x$

Soln : General solution is $y = c_1 \cos 5x + c_2 \sin 5x$

Here $\alpha = 0$ and $\beta = 5$

$\therefore$ Roots of the A.E are $\alpha \pm i\beta = \pm 5i$

A.E is $(D-5i)(D+5i)=0 \Rightarrow D^2+25=0$

$\therefore$ The equation is $(D^2+25)y=0$ or

$$\frac{d^2 y}{dx^2} + 25y = 0$$

iii) $e^{-2.5x} \cos 0.5x$, $e^{-2.5x} \sin 0.5x$

Soln : General solution is $y = e^{-2.5x} (c_1 \cos 0.5x + c_2 \sin 0.5x)$

Here $\alpha = -2.5$ and $\beta = 0.5$

$\therefore$ Roots of the A.E are $-2.5 \pm i(0.5)$

(12)

$$\therefore \text{A.E. is } [D - (-2.5 + i0.5)][D - (-2.5 - i0.5)] = 0$$

$$\Rightarrow [(D + 2.5) - i0.5][(D + 2.5) + i0.5] = 0$$

$$\Rightarrow (D + 2.5)^2 + (0.5)^2 = 0$$

$$\Rightarrow D^2 + 5D + 6.5 = 0$$

$$\therefore \text{The eqn is } (D^2 + 5D + 6.5)y = 0 \text{ or}$$

$$\underline{\underline{\frac{d^2 y}{dx^2} + 5 \frac{dy}{dx} + 6.5y = 0}}}$$

Problems

Solve the following differential equations

1) $\frac{d^2 y}{dx^2} - 3 \frac{dy}{dx} - 4y = 0$

4) $\frac{d^3 y}{dx^3} + 6 \frac{d^2 y}{dx^2} + 11 \frac{dy}{dx} + 6y = 0$

2) $y'' + (a+b)y' + aby = 0$

5) $y^{IV} - 5y'' + 4y = 0$

3) $\frac{d^2 y}{dx^2} - 4 \frac{dy}{dx} + y = 0$

6) $\frac{d^4 y}{dx^4} + 8 \frac{d^2 y}{dx^2} + 16y = 0$

7) $\frac{d^2 x}{dt^2} - 3 \frac{dx}{dt} + 2x = 0$; $x(0) = 0$; $x'(0) = 0$

8) If $\frac{d^4 x}{dt^4} = m^4 x$, show that

$$x = c_1 \cos mt + c_2 \sin mt + c_3 \cosh mt + c_4 \sinh mt$$

(13)

To solve a non homogeneous linear ODE with constant coefficients

Consider the non homogeneous linear ODE

$f(D)y = x$ where x is either a constant or any function of x

If $y = y_h$ is the general solution of $f(D)y = 0$

and $y = y_p$ is a particular solution of $f(D)y = x$, then the general solution of $f(D)y = x$ is

$$y = y_h + y_p.$$

y_h is called complementary function and

y_p is called particular integral.

To find the particular integral

- 1) Method of undetermined coefficients
- 2) Method of variation of parameters

1) Method of undetermined coefficients

This method is suitable for linear ODE with constant coefficients $y'' + ay' + by = x$ where x is an exponential function, a cosine or sine function, a power of x or sum or product of such functions. (Because, these functions have derivative similar to x itself)

To find the particular integral, we choose a trial solution containing unknown constants which are determined by substitution in the given equation.

(14)

X

$$k e^{ax}$$

$$k x^n$$

$$k \cos ax$$

$$k \sin ax$$

$$k e^{ax} \cos bx$$

$$k e^{ax} \sin bx$$

Trial Solution

$$A e^{ax}$$

$$A + Bx + Cx^2 + \dots + Nx^n$$

$$A \cos ax + B \sin ax$$

„

$$e^{ax} (A \cos bx + B \sin bx)$$

„

1. Solve $y'' + 5y' + 6y = 2e^{-x}$

Soln : A.E is $(D^2 + 5D + 6) = 0$

$$\Rightarrow (D+3)(D+2) = 0 \Rightarrow D = -3, -2 \text{ (real \& distinct)}$$

Complementary function, $y_h = c_1 e^{-3x} + c_2 e^{-2x}$

Let the particular integral be $y_p = A e^{-x}$

$$\therefore y_p' = -A e^{-x} ; y_p'' = A e^{-x}$$

Substituting in the given equation,

$$A e^{-x} + 5(-A e^{-x}) + 6A e^{-x} = 2e^{-x}$$

$$\Rightarrow 2A e^{-x} = 2e^{-x} \Rightarrow 2A = 2 \Rightarrow A = 1$$

$$\therefore y_p = e^{-x}$$

$$\therefore \text{General solution is } y = y_h + y_p = \underline{\underline{c_1 e^{-3x} + c_2 e^{-2x} + e^{-x}}}$$

(15)

2. Solve $\frac{d^2 y}{dx^2} - y = 3 \cos x$

Soln : A.E is $D^2 - 1 = 0 \Rightarrow D = \pm 1$ (real & distinct)

$$\therefore y_h = c_1 e^x + c_2 e^{-x}$$

Let $y_p = A \cos x + B \sin x$

$$y_p' = -A \sin x + B \cos x ; y_p'' = -A \cos x - B \sin x$$

Substituting in the given equation,

$$-A \cos x - B \sin x - A \cos x - B \sin x = 3 \cos x$$

$$\Rightarrow -2A \cos x - 2B \sin x = 3 \cos x$$

Comparing the coefficients,

$$-2A = 3 \quad \text{and} \quad -2B = 0$$

$$\Rightarrow A = -\frac{3}{2} \quad B = 0$$

$$\therefore y_p = -\frac{3}{2} \cos x$$

$\therefore$ General solution is $y = y_h + y_p = c_1 e^x + c_2 e^{-x} - \frac{3}{2} \cos x$

3. Solve $y'' + 3y' + 2y = 12x^2$

Solution : A.E is $D^2 + 3D + 2 = 0$

$$\Rightarrow (D+2)(D+1) = 0 \Rightarrow D = -2, -1$$

$$\therefore y_h = c_1 e^{-2x} + c_2 e^{-x}$$

Let $y_p = A + Bx + Cx^2$

$$y_p' = B + 2Cx ; y_p'' = 2C$$

Substituting in the given equation,

$$2C + 3(B + 2Cx) + 2(A + Bx + Cx^2) = 12x^2$$

$$\Rightarrow (2C + 3B + 2A) + (6C + 2B)x + 2Cx^2 = 12x^2$$

(16)

Comparing the coefficients,

$$2C = 12$$

$$\Rightarrow C = 6$$

$$6C + 2B = 0$$

$$6(6) + 2B = 0$$

$$\Rightarrow B = -18$$

$$2C + 3B + 2A = 0$$

$$2(6) + 3(-18) + 2A = 0$$

$$\Rightarrow A = 21$$

$$\therefore y_p = 21 - 18x + 6x^2$$

$$\therefore y = y_h + y_p = \underline{\underline{C_1 e^{-2x} + C_2 e^{-x} + 6x^2 - 18x + 21}}$$

4. Solve $y'' + 4y' + 4y = e^{-x} \cos x$

Soln:- A.E is $D^2 + 4D + 4 = 0$

$$\Rightarrow (D+2)(D+2) = 0 \Rightarrow D = -2, -2 \text{ (equal roots)}$$

$$\therefore y_h = (C_1 x + C_2) e^{-2x}$$

Let $y_p = e^{-x} (A \cos x + B \sin x) = A e^{-x} \cos x + B e^{-x} \sin x$

$$y_p' = -A e^{-x} \sin x - A e^{-x} \cos x + B e^{-x} \cos x - B e^{-x} \sin x$$

$$\begin{aligned} y_p'' &= -A e^{-x} \cos x + A e^{-x} \sin x + A e^{-x} \sin x + A e^{-x} \cos x \\ &\quad - B e^{-x} \sin x - B e^{-x} \cos x - B e^{-x} \cos x + B e^{-x} \sin x \\ &= 2A e^{-x} \sin x - 2B e^{-x} \cos x \end{aligned}$$

Substituting in the given equation,

$$2A e^{-x} \sin x - 2B e^{-x} \cos x + 4(-A e^{-x} \sin x - A e^{-x} \cos x + B e^{-x} \cos x - B e^{-x} \sin x)$$

$$+ 4(A e^{-x} \cos x + B e^{-x} \sin x) = e^{-x} \cos x$$

$$\Rightarrow (2A - 4A - 4B + 4B) e^{-x} \sin x + (-2B - 4A + 4B + 4A) e^{-x} \cos x = e^{-x} \cos x$$

$$\Rightarrow (-2A) e^{-x} \sin x + (2B) e^{-x} \cos x = e^{-x} \cos x$$

(17)

Comparing the coefficients,

$$-2A = 0 \quad \text{and} \quad 2B = 1$$

$$\Rightarrow A = 0 \quad \text{and} \quad B = \frac{1}{2}$$

$$\therefore y_p = \frac{1}{2} e^{-x} \sin x$$

$$\therefore y = y_h + y_p = \underline{\underline{(C_1 x + C_2) e^{-x} + \frac{1}{2} e^{-x} \sin x}}$$

5. Solve $y'' + 2y' + 4y = 2x^2 + 3e^{-x}$

Soln: A.E is $D^2 + 2D + 4 = 0$

$$\Rightarrow D = \frac{-2 \pm \sqrt{4 - 16}}{2} = \frac{-2 \pm i2\sqrt{3}}{2}$$

$$= -1 \pm i\sqrt{3} \quad (\text{complex roots})$$

$$\therefore y_h = e^{-x} (C_1 \cos \sqrt{3} x + C_2 \sin \sqrt{3} x)$$

Let $y_p = (A + Bx + Cx^2) + D e^{-x}$

$$y_p' = B + 2Cx - D e^{-x} \quad ; \quad y_p'' = 2C + D e^{-x}$$

Substituting in the given equation,

$$2C + D e^{-x} + 2(B + 2Cx - D e^{-x}) + 4(A + Bx + Cx^2 + D e^{-x}) = 2x^2 + 3e^{-x}$$

$$\Rightarrow (2C + 2B + 4A) + (4C + 4B)x + (4C)x^2 + (D - 2D + 4D)e^{-x} = 2x^2 + 3e^{-x}$$

Comparing the coefficients,

$$2C + 2B + 4A = 0 \quad ; \quad 4C + 4B = 0 \quad ; \quad 4C = 2 \quad ; \quad 3D = 3$$

$$\Rightarrow C = \frac{1}{2}$$

$$\Rightarrow D = 1$$

$$\Rightarrow 4\left(\frac{1}{2}\right) + 4B = 0$$

$$\Rightarrow B = -\frac{1}{2}$$

(18)

$$2C + 2B + 4A = 0 \Rightarrow 2\left(\frac{1}{2}\right) + 2\left(\frac{-1}{2}\right) + 4A = 0$$
$$\Rightarrow 4A = 0 \Rightarrow A = 0$$

$$\therefore y_p = \frac{-1}{2}x + \frac{1}{2}x^2 + e^{-x}$$

$$\therefore y = y_h + y_p = e^{-x}(c_1 \cos \sqrt{3}x + c_2 \sin \sqrt{3}x) + \left(\frac{-1}{2}x + \frac{1}{2}x^2 + e^{-x}\right)$$

Modification rule

If any term in the trial solution appears in the complementary function, we multiply this trial solution by x . If this solution corresponds to a double root of A.E, multiply by x^2 and so on.

6. Solve $(D^2 + 1)y = \sin x$

Soln: A.E is $D^2 + 1 = 0 \Rightarrow D = \pm i$

$$\therefore y_h = c_1 \cos x + c_2 \sin x$$

Let $y_p = (A \cos x + B \sin x)x = Ax \cos x + Bx \sin x$

$$y_p' = -Ax \sin x + A \cos x + Bx \cos x + B \sin x$$

$$y_p'' = -Ax \cos x - A \sin x - A \sin x - Bx \sin x + B \cos x + B \cos x$$

$$= -Ax \cos x - 2A \sin x - Bx \sin x + 2B \cos x$$

Substituting in the given equation,

$$-Ax \cancel{\cos x} - 2A \sin x - Bx \cancel{\sin x} + 2B \cos x + A \cancel{\cos x} + B \cancel{\sin x} = \sin x$$

$$\Rightarrow -2A \sin x + 2B \cos x = \sin x$$

(19)

Comparing the coefficients,

$$-2A = 1 \quad \text{and} \quad 2B = 0$$

$$\Rightarrow A = -\frac{1}{2}$$

$$B = 0$$

$$\therefore y_p = -\frac{1}{2} x \cos x$$

$$\therefore y = y_h + y_p = \underline{\underline{C_1 \cos x + C_2 \sin x - \frac{1}{2} x \cos x}}$$

7. Solve $y'' + 2y' + y = e^{-x}$

Soln: A.E is $(D^2 + 2D + 1) = 0 \Rightarrow (D+1)^2 = 0 \Rightarrow D = -1, -1$
(equal roots)

$$y_h = (C_1 x + C_2) e^{-x}$$

Let $y_p = A e^{-x} (x^2)$

$$y_p' = -A x^2 e^{-x} + 2A x e^{-x}$$

$$y_p'' = A x^2 e^{-x} - 2A x e^{-x} - 2A x e^{-x} + 2A e^{-x} = A x^2 e^{-x} - 4A x e^{-x} + 2A e^{-x}$$

Substituting in the given equation,

$$(A x^2 e^{-x} - 4A x e^{-x} + 2A e^{-x}) + 2(-A x^2 e^{-x} + 2A x e^{-x}) + A x^2 e^{-x} = e^{-x}$$

$$\Rightarrow 2A e^{-x} = e^{-x} \Rightarrow 2A = 1 \Rightarrow A = \frac{1}{2}$$

$$\therefore y_p = \frac{1}{2} e^{-x} x^2$$

$$\therefore y = y_h + y_p = \underline{\underline{(C_1 x + C_2) e^{-x} + \frac{1}{2} e^{-x} x^2}}$$

8. Solve $\frac{d^2 y}{dx^2} + \frac{dy}{dx} = x^2 + 2x + 4$

Soln: A.E is $D^2 + D = 0 \Rightarrow D(D+1) = 0 \Rightarrow D = 0, -1$

(20)

$$y_h = c_1 e^{0x} + c_2 e^{-x} = c_1 + c_2 e^{-x}$$

$$\text{Let } y_p = (A + Bx + Cx^2)x = Ax + Bx^2 + Cx^3$$

$$y_p' = A + 2Bx + 3Cx^2 \quad ; \quad y_p'' = 2B + 6Cx$$

Substituting in the given equation,

$$(2B + 6Cx) + (A + 2Bx + 3Cx^2) = x^2 + 2x + 4$$

$$3Cx^2 + (6C + 2B)x + (A + 2B) = x^2 + 2x + 4$$

Comparing the coefficients,

$$3C = 1$$

$$6C + 2B = 2$$

$$A + 2B = 4$$

$$\Rightarrow C = \frac{1}{3}$$

$$6\left(\frac{1}{3}\right) + 2B = 2$$

$$A + 2(0) = 4$$

$$\Rightarrow B = 0$$

$$\Rightarrow A = 4$$

$$\therefore y_p = 4x + \frac{1}{3}x^3$$

$$\therefore y = c_1 + c_2 e^{-x} + 4x + \frac{x^3}{3}$$

9. Solve $\frac{d^2 y}{dx^2} + 4y = x \sin x$

Soln : A.E is $D^2 + 4 = 0 \Rightarrow D^2 = -4 \Rightarrow D = \pm 2i$

$$\therefore y_h = c_1 \cos 2x + c_2 \sin 2x$$

$$\text{Let } y_p = (A + Bx)(C \cos x + D \sin x)$$

$$= AC \cos x + AD \sin x + BCx \cos x + BDx \sin x$$

$$y_p' = -AC \sin x + AD \cos x - BCx \sin x + BC \cos x + BDx \cos x + BD \sin x$$

$$y_p'' = -AC \cos x - AD \sin x - BCx \cos x - BC \sin x - BC \sin x - BDx \sin x + BD \cos x + BD \cos x$$

$$= (-AC + 2BD) \cos x + (-AD - 2BC) \sin x - BCx \cos x - BDx \sin x$$

(21)

Substituting in the given equation,

$$(-AC + 2BD)\cos x + (-AD - 2BC)\sin x - BCx\cos x - BDx\sin x \\ + 4(AC\cos x + AD\sin x + BCx\cos x + BDx\sin x) = x\sin x$$

$$\Rightarrow (3AC + 2BD)\cos x + (3AD - 2BC)\sin x + 3BCx\cos x + 3BDx\sin x = x\sin x$$

Comparing the coefficients,

$$3AC + 2BD = 0 \quad ; \quad 3AD - 2BC = 0 \quad ; \quad 3BC = 0 \quad ; \quad 3BD = 1 \\ \Rightarrow BC = 0 \quad \Rightarrow BD = \frac{1}{3}$$

$$\Rightarrow 3AC + 2\left(\frac{1}{3}\right) = 0$$

$$\Rightarrow 3AD - 2(0) = 0$$

$$\Rightarrow AC = -\frac{2}{9}$$

$$\Rightarrow AD = 0$$

$$\therefore y_p = -\frac{2}{9}\cos x + \frac{1}{3}x\sin x$$

$$\therefore y = y_h + y_p = \underline{c_1 \cos 2x + c_2 \sin 2x} - \frac{2}{9}\cos x + \frac{x\sin x}{3}$$

Problems :-Solve the following differential equations

1) $\frac{d^3 y}{dx^3} + y = 3 + 5e^x$

2) $y'' - 2y' + 5y = \sin 3x$

3) $y'' - 3y' + 2y = 6e^{-3x} + \sin 2x$

4) $\frac{d^3 y}{dx^3} - \frac{d^2 y}{dx^2} - 6\frac{dy}{dx} = 1 + x^2$

5) $\frac{d^4 y}{dx^4} - y = e^x \cos x$

6) $\frac{d^2 y}{dx^2} - 4y = (1 + e^x)^2$

7) $(D^2 - 4D + 3)y = \sin 3x \cos 2x$

8) $\frac{d^2 y}{dx^2} - 4y = x^2$

(22)

9) $y'' + y = 2 \cos x$

10) $y'' + y = x \sin x$

11) $\frac{d^3 y}{dx^3} + \frac{d^2 y}{dx^2} + \frac{dy}{dx} + y = \sin 2x$

12) $y'' + 4y = e^x + \sin 2x$

13) $\frac{d^2 y}{dx^2} + 3 \frac{dy}{dx} + 2y = x e^{3x} + \sin 2x$

2) Method of variation of parameters to find the particular integral

Consider a second order linear ODE with constant coefficients,

$$y'' + ay' + by = X$$

Let the complementary function be,

$$y_h = c_1 y_1 + c_2 y_2$$

Then P.I $y_p = u(x)y_1 + v(x)y_2$ where

$$u(x) = - \int \frac{y_2 X}{W} dx \quad \text{and} \quad v(x) = \int \frac{y_1 X}{W} dx$$

where W is the Wronskian of y_1 and y_2 .

1. Solve $y'' + y = \operatorname{cosec} x$

Soln :- A.E is $D^2 + 1 = 0 \Rightarrow D = \pm i$

C.F is $y_h = c_1 \cos x + c_2 \sin x$

Here $y_1 = \cos x$; $y_2 = \sin x$ and $X = \operatorname{cosec} x$

(23)

$$\text{Wronskian, } W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} \\ = \cos^2 x + \sin^2 x = 1$$

$$\therefore u(x) = - \int \frac{y_2 x}{W} dx = - \int \sin x \cos x dx = - \int dx = \underline{-x}$$

$$v(x) = \int \frac{y_1 x}{W} dx = \int \cos x \cos x dx = \int \frac{\cos x}{\sin x} dx \\ = \int \frac{1}{u} du \quad \left(\begin{array}{l} \text{Put } \sin x = u \\ \cos x dx = du \end{array} \right) \\ = \log u = \log(\sin x)$$

$$\therefore \text{P.I.}, y_p = u(x)y_1 + v(x)y_2 = \underline{-x \cos x + \sin x \log(\sin x)}$$

$$\therefore y = y_h + y_p = \underline{c_1 \cos x + c_2 \sin x - x \cos x + \sin x \log(\sin x)}$$

2. Solve $y'' + 4y = 4 \sec^2 2x$

Soln : A.E is $D^2 + 4 = 0 \Rightarrow D = \pm 2x$

C.F, $y_h = c_1 \cos 2x + c_2 \sin 2x$

Here $y_1 = \cos 2x$; $y_2 = \sin 2x$; $x = 4 \sec^2 2x$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} \cos 2x & \sin 2x \\ -2 \sin 2x & 2 \cos 2x \end{vmatrix} = 2 \cos^2 2x + 2 \sin^2 2x \\ = \underline{2}$$

$$u(x) = - \int \frac{y_2 x}{W} dx = - \int \frac{\sin 2x \cdot 4 \sec^2 2x}{2} dx \\ = -2 \int \frac{\sin 2x}{\cos^2 2x} dx = -2 \int \sec 2x \tan 2x dx \\ = -2 \frac{\sec 2x}{2} = \underline{-\sec 2x}$$

(24)

$$\begin{aligned}
 v(x) &= \int \frac{y_1 x}{w} dx = \int \frac{\cos 2x \cdot \sec^2 2x}{2} dx \\
 &= 2 \int \sec 2x dx \\
 &= 2 \frac{\log(\sec 2x + \tan 2x)}{2} \\
 &= \log(\sec 2x + \tan 2x)
 \end{aligned}
 \quad \left[\begin{aligned} \int \sec u dx &= \frac{\sec u (\sec u + \tan u)}{\sec u + \tan u} \\ &= \log(\sec u + \tan u) \end{aligned} \right]$$

$$\begin{aligned}
 \therefore y_p &= u(x)y_1 + v(x)y_2 = -\sec 2x \cos 2x + \sin 2x \log(\sec 2x + \tan 2x) \\
 &= -1 + \sin 2x \log(\sec 2x + \tan 2x)
 \end{aligned}$$

$$\therefore y = y_h + y_p = c_1 \cos 2x + c_2 \sin 2x - 1 + \sin 2x \log(\sec 2x + \tan 2x)$$

3. Solve $y'' + 4y = \tan 2x$

Soln :- A.E is $D^2 + 4 = 0 \Rightarrow D = \pm 2i$

$$\text{C.F. , } y_h = c_1 \cos 2x + c_2 \sin 2x$$

$$\text{Here , } y_1 = \cos 2x, \quad y_2 = \sin 2x \quad x = \tan 2x$$

$$w = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} \cos 2x & \sin 2x \\ -2\sin 2x & 2\cos 2x \end{vmatrix} = 2$$

$$\begin{aligned}
 u(x) &= - \int \frac{y_2 x}{w} dx = - \int \frac{\sin 2x \tan 2x}{2} dx = -\frac{1}{2} \int \frac{\sin^2 2x}{\cos 2x} dx \\
 &= -\frac{1}{2} \int \frac{1 - \cos^2 2x}{\cos 2x} dx \\
 &= -\frac{1}{2} \int (\sec 2x - \cos 2x) dx \\
 &= -\frac{1}{2} \left[\frac{\log(\sec 2x + \tan 2x)}{2} - \frac{\sin 2x}{2} \right]
 \end{aligned}$$

(25)

$$v(x) = \int \frac{y_1 x}{w} dx = \int \frac{\cos 2x \tan 2x}{2} dx = \frac{1}{2} \int \sin 2x dx$$

$$= -\frac{\cos 2x}{4}$$

$$\therefore y_p = u(x)y_1 + v(x)y_2 = \frac{-\cos 2x \log(\sec 2x + \tan 2x)}{4} - \frac{\sin 2x \cos 2x}{4}$$

$$\therefore y = y_h + y_p = c_1 \cos 2x + c_2 \sin 2x - \frac{\cos 2x \log(\sec 2x + \tan 2x)}{4} - \frac{\sin 2x \cos 2x}{4}$$

4. Solve $y'' - 2y' + 2y = e^x \cdot \tan x$

Soln:- A.E is $D^2 - 2D + 2 = 0 \Rightarrow D = 1 \pm i$

$$\therefore y_h = e^x (c_1 \cos x + c_2 \sin x)$$

Here $y_1 = e^x \cos x$; $y_2 = e^x \sin x$; $X = e^x \tan x$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^x \cos x & e^x \sin x \\ -e^x \sin x + e^x \cos x & e^x \cos x + e^x \sin x \end{vmatrix}$$

$$= e^{2x} \cos^2 x + e^{2x} \cos x \sin x + e^{2x} \sin^2 x - e^{2x} \sin x \cos x$$

$$= \underline{e^{2x}}$$

$$u(x) = - \int \frac{y_2 X}{w} dx = - \int \frac{e^x \sin x e^x \tan x}{e^{2x}} dx$$

$$= - \int \frac{\sin^2 x}{\cos x} dx = - \int \frac{1 - \cos^2 x}{\cos x} dx$$

$$= - \int (\sec x - \cos x) dx$$

$$= - \log(\sec x + \tan x) + \sin x$$

$$v(x) = \int \frac{y_1 X}{w} dx = \int \frac{e^x \cos x e^x \tan x}{e^{2x}} dx = \int \sin x dx = -\cos x$$

(26)

$$\therefore y_p = u(x)y_1 + v(x)y_2 = -e^x \cos x [\log(\sec x + \tan x) - \sin x] - e^x \sin x \cos x$$

$$\therefore y = y_h + y_p = e^x (c_1 \cos x + c_2 \sin x) - e^x \cos x [\log(\sec x + \tan x) - \sin x] - e^x \sin x \cos x$$

Problems

Solve using method of variation of parameters

i) $\frac{d^2 y}{dx^2} + y = \sec x$

ii) $\frac{d^2 y}{dx^2} + y = x \sin x$

Euler-Cauchy Equation

An equation of the form

$$x^n \frac{d^n y}{dx^n} + a_1 x^{n-1} \frac{d^{n-1} y}{dx^{n-1}} + a_2 x^{n-2} \frac{d^{n-2} y}{dx^{n-2}} + \dots + a_{n-1} x \frac{dy}{dx} + a_n y = x$$

where a_i 's are constants and x is a function of x , is called Euler Cauchy's equation.

Such equations can be reduced to linear differential equations with constant coefficients by the substitution, $x = e^z$ or $z = \log x$

$$\frac{dy}{dx} = \frac{dy}{dz} \cdot \frac{dz}{dx} = \frac{dy}{dz} \cdot \frac{1}{x}$$

$$\Rightarrow x \frac{dy}{dx} = \frac{dy}{dz} = Dy \quad \text{where } D = \frac{d}{dz}$$

$$\text{Similarly, } x^2 \frac{d^2 y}{dx^2} = D(D-1)y; \quad x^3 \frac{d^3 y}{dx^3} = D(D-1)(D-2) \text{ etc.}$$

(27)

1. Solve $x^2 y'' - 5xy' + 9y = 0$

Soln :- The given equation is an Euler-Cauchy equation.

Put $x = e^z \Rightarrow z = \log x$

$x \frac{dy}{dx} = Dy$ where $D = \frac{d}{dz}$

$x^2 \frac{d^2 y}{dx^2} = D(D-1)y$

Substituting in the given equation,

$$[D(D-1) - 5D + 9]y = 0$$

$$\Rightarrow [D^2 - 6D + 9]y = 0$$

A.E is $D^2 - 6D + 9 = 0 \Rightarrow D = 3, 3$

$$\therefore y = (c_1 z + c_2) e^{3z} = \underline{\underline{(c_1 \log x + c_2) x^3}}$$

2. Solve $4x^2 y'' + 5y = 0$

Soln :- Put $x = e^z \Rightarrow z = \log x$

$x^2 \frac{d^2 y}{dx^2} = D(D-1)y$ where $D = \frac{d}{dz}$

Substituting in the given equation,

$$[4D(D-1) + 5]y = 0$$

A.E is $4D^2 - 4D + 5 = 0$

$$\Rightarrow D = \frac{4 \pm \sqrt{16 - 80}}{8} = \frac{1}{2} \pm i$$

$$\therefore y = e^{\frac{z}{2}} (c_1 \cos z + c_2 \sin z)$$

$$= \sqrt{x} \left[\underline{\underline{c_1 \cos(\log x) + c_2 \sin(\log x)}} \right]$$

(28)

3. Solve $x^2 \frac{d^2 y}{dx^2} - x \frac{dy}{dx} + y = \log x$

Soln :- Put $x = e^z \Rightarrow z = \log x$

$$x \frac{dy}{dx} = Dy \text{ where } D = \frac{d}{dz}$$

$$x^2 \frac{d^2 y}{dx^2} = D(D-1)y$$

Substituting in the given equation,

$$[D(D-1) - D + 1]y = z \text{ ——— (1)}$$

A.E is $D^2 - 2D + 1 = 0 \Rightarrow D = 1, 1$

$$\therefore \text{C.F, } y_h = (c_1 z + c_2) e^z = (c_1 \log x + c_2)x$$

Let $y_p = A + Bz$

$$y_p' = B ; y_p'' = 0$$

Substituting in eq(1)

$$0 - 2B + A + Bz = z \Rightarrow B = 1 \text{ and } -2B + A = 0 \\ \Rightarrow -2 + A = 0 \\ \Rightarrow A = 2$$

$$\therefore y_p = 2 + z = 2 + \log x$$

$$\therefore y = y_h + y_p = \underline{\underline{(c_1 \log x + c_2)x + 2 + \log x}}$$

4. Solve $x^3 \frac{d^3 y}{dx^3} + 2x^2 \frac{d^2 y}{dx^2} + 2y = 10(x + \frac{1}{x})$

Ans Put $x = e^z \Rightarrow z = \log x$

$$x \frac{dy}{dx} = Dy \text{ where } D = \frac{d}{dz} ; x^2 \frac{d^2 y}{dx^2} = D(D-1)y$$

$$x^3 \frac{d^3 y}{dx^3} = D(D-1)(D-2)y$$

(29)

Substituting in the given equation,

$$[D(D-1)(D-2) + 2D(D-1) + 2]y = 10(e^x + e^{-x})$$

$$\Rightarrow (D^3 - D^2 + 2)y = 10(e^x + e^{-x}) \quad \text{--- (1)}$$

A.E. is $D^3 - D^2 + 2 = 0 \Rightarrow D = -1, 1 \pm i$

$$\therefore \text{C.F.}, y_h = c_1 e^{-x} + e^x (c_2 \cos x + c_3 \sin x)$$

$$= \frac{c_1}{x} + x [c_2 \cos(\log x) + c_3 \sin(\log x)]$$

Let $y_p = Ae^x + Be^{-x}$

$$y_p' = Ae^x - B_1 e^{-x} + B e^{-x}$$

$$y_p'' = Ae^x + B_1 e^{-x} - B e^{-x} - B e^{-x} = Ae^x + B_1 e^{-x} - 2B e^{-x}$$

$$y_p''' = Ae^x - B_1 e^{-x} + B e^{-x} + 2B e^{-x} = Ae^x - B_1 e^{-x} + 3B e^{-x}$$

Substituting in eq(1)

$$\cancel{Ae^x} - \cancel{B_1 e^{-x}} + 3B e^{-x} - \cancel{Ae^x} - \cancel{B_1 e^{-x}} + 2B e^{-x} + 2Ae^x + 2B_1 e^{-x}$$

$$= 10e^x + 10e^{-x}$$

$$\Rightarrow 2Ae^x + 5B e^{-x} = 10e^x + 10e^{-x}$$

$$\Rightarrow 2A = 10 \quad \text{and} \quad 5B = 10$$

$$\Rightarrow A = 5 \quad B = 2$$

$$\therefore y_p = 5e^x + 2e^{-x} = 5x + \frac{2 \log x}{x}$$

$$\therefore y = y_h + y_p = \frac{c_1}{x} + x [c_2 \cos(\log x) + c_3 \sin(\log x)] + 5x + \frac{2 \log x}{x}$$

(30)

Problems

Solve

$$1) \quad x^2 \frac{d^2 y}{dx^2} + 9x \frac{dy}{dx} + 25y = 50$$

$$2) \quad x^2 \frac{d^2 y}{dx^2} - 2y = x^2 + \frac{1}{x}$$

$$3) \quad x^2 \frac{d^2 y}{dx^2} + 2x \frac{dy}{dx} - 20y = (x+1)^2$$

$$4) \quad x^2 \frac{d^3 y}{dx^3} - 4x \frac{d^2 y}{dx^2} + 6 \frac{dy}{dx} = 4 \quad [\text{Hint: Multiply throughout by } x]$$

$$5) \quad x^4 \frac{d^3 y}{dx^3} + 2x^3 \frac{d^2 y}{dx^2} - x^2 \frac{dy}{dx} + xy = 1$$

$$6) \quad x^2 \frac{d^2 y}{dx^2} - 2x \frac{dy}{dx} - 4y = x^2 + 2 \log x$$

$$7) \quad x^2 \frac{d^2 y}{dx^2} - 3x \frac{dy}{dx} + 5y = \sin(\log x)$$